Peng 2019 PDAC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Pre-Harmony the 16 tumours form a per-patient archipelago (iLISI median 1.0 of 16, many detached islands). With Harmony on sample they intermix into one continuous mass — iLISI median 4.05, a single kNN-graph component, 0 detached cells. Both mixed and continuous. θ=2 from a mixing+geometry sweep (mixing saturates by θ=2; higher θ only collapses the developmental axis, aspect 1.37→1.05).

Gate 2 — module geography → GO (non-branching)

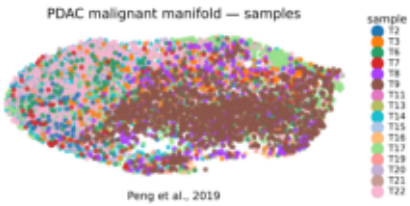
Decided on geography, not means. Not a Puram hub-and-spoke: no Gast/pSom co-expression hub (interior Gast-XOR vs pSom-XOR r=+0.29, not −0.49) and a PPT tree over-fits the continuous cloud (26–43 forks, no clean Y). Non-branching case (§1): organized Gast↔pSom cross-repression (spatial r=−0.53) with a terminal neural-crest gradient — pSom↔Crest r=−0.57 (GO range: Steele −0.68, Moncada −0.56), and on the origin-rooted pseudotime Crest is the latest/terminal module (ρ=+0.40). Crest opposes pSom more than Gast — no Gast-arm inversion (no flag).

Gate 3 — hub cross-repression → NA

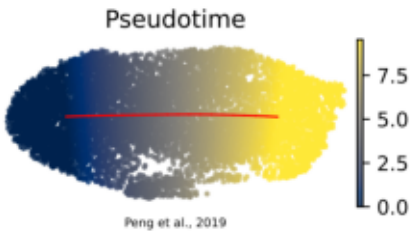
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free). With no bifurcation there is no Mixed-type hub and no branch pair to test for mutual Gast↔pSom exclusivity, so the branching-only 04_correlation / 05_modules notebooks are not produced (per the deliverable spec).

STEP FIGURES

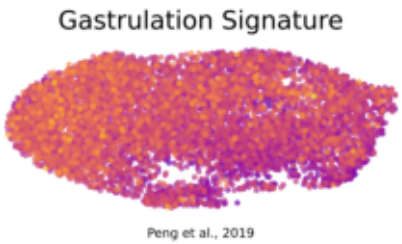
Manifold — samples



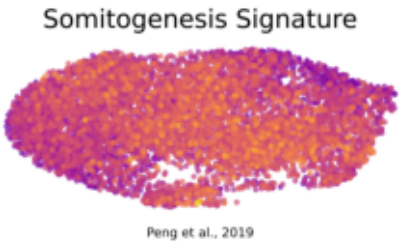
Pseudotime



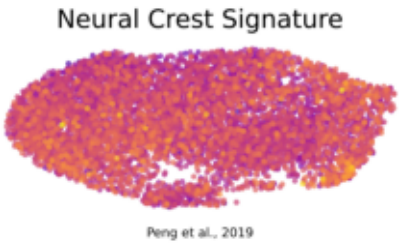
Gastrulation



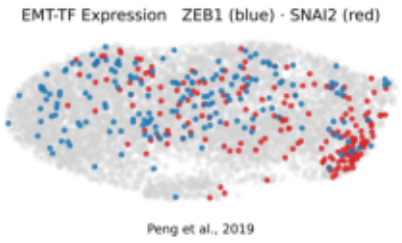
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Peng et al., 2019), 10x — 16 tumours / 27,953 cells (master scope); malignant only: 7,945 cells across all 16 samples (≥100-cell floor; the multiplicative count gap is only 2.31×, mid-distribution, so no clean spillover cliff). Whole malignant manifold (two density lobes, near-identical developmental means, not scoped).

Verdict.
Pattern reproduces in the non-branching form (Linear). One continuous, sample-integrated manifold (G1 GO) carrying organized Gast↔pSom cross-repression with a terminal neural-crest gradient (G2 GO): rooted at the gastrulation/somitogenesis-high origin, the developmental ordering is Gast → pSom → Crest-terminal, matching the passing same-tissue Steele/Moncada PDAC runs. It lacks Puram's hub-and-spoke bifurcation, so there is no hub to test (G3 NA). EMT-TFs ZEB1 (root-side) → SNAI2 (far end); canonical SNAI1/TWIST near the detection floor.

Parameter choices

Sample threshold	≥100-cell floor (mult. gap 2.31× < 4× degenerate); 16/16 kept → 7,945 cells	**Overdispersed genes**	1,288 (scFates find_overdispersed); no sc.pp.scale before PCA
Integration	Harmony on sample, θ=2 (mixing saturates by θ=2; higher collapses axis 1.37→1.05)	**Embedding**	n_pcs=20, n_neighbors=30, UMAP spread=1.3; 0 islands removed; 1 component
Geometry	Linear (non-branching) principal curve; no hub, organized cross-repression + terminal Crest	**Root**	tip 0 = Gast/pSom-high, Crest-low origin; curve fold-free (ratio 2.98); Crest terminal on DPT
Module means (raw)	Gast −0.047 · pSom +0.098 · Crest −0.055 (means match Steele PASS; geography decides)	**EMT-TFs shown**	ZEB1 (8.8%) & SNAI2 (9.3%) — most end-separated expressed factors; VIM excluded (not a TF)